

STAVELY AAFC, AB, Canada

WMO: 715550

Lat: 50.183N

Lon: 113.883W

Elev: 4474

StdP: 12.47

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.7	-9.7	-22.1	1.9	-13.5	-15.8	2.7	-6.7	35.9	37.8	31.1	36.5	7.2	340	0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	81.5	58.8	78.1	57.7	74.7	56.8	62.7	74.2	60.8	73.0	58.9	71.2	11.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	86.2	68.5	56.1	79.1	66.2	53.8	72.7	64.2	30.7	74.3	29.1	72.8	27.7	71.1	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.2	26.8	22.9	DB	-19.9	87.7	6.2	3.8	-24.4	90.5	-28.0	92.7	-31.5	94.9	-36.0	97.7	(4)
(5)			WB	-20.6	66.7	6.2	2.9	-25.0	68.8	-28.6	70.4	-32.1	72.1	-36.6	74.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.3	24.7	23.9	30.5	38.7	48.3	54.6	61.9	61.0	53.0	42.2	31.8	24.2
	DBStd	17.57	15.28	13.65	12.85	10.00	8.96	6.44	6.49	7.21	9.80	10.09	12.23	14.01
	HDD50	4362	787	730	607	357	140	25	3	8	78	278	549	800
	HDD65	8737	1251	1150	1071	790	519	318	138	162	372	707	995	1264
	CDD50	1202	1	0	1	17	88	163	372	350	169	36	4	1
	CDD65	99	0	0	0	0	2	5	41	38	13	0	0	0
	CDH74	851	0	0	0	1	21	38	344	339	107	1	0	0
	CDH80	163	0	0	0	0	1	5	70	70	17	0	0	0
Wind	WSAvg	10.2	10.9	9.6	11.0	10.9	10.4	9.9	9.4	8.8	9.4	10.4	11.4	11.0
Precipitation	PrecAvg	18.4	0.6	0.7	0.9	1.5	2.6	4.1	1.9	2.0	1.8	0.9	0.8	0.7
	PrecMax	31.9	1.1	1.5	1.6	2.8	6.1	14.6	4.6	4.5	5.5	3.0	1.7	1.4
	PrecMin	9.2	0.2	0.2	0.2	0.6	0.2	1.0	0.0	0.0	0.2	0.2	0.2	0.2
	PrecStd	4.9	0.2	0.3	0.4	0.6	1.5	3.1	1.3	1.1	1.3	0.7	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	53.1	58.6	69.5	77.0	79.3	86.1	86.0	82.3	70.4	60.8	53.9
		MCWB	39.1	37.9	42.5	47.4	53.3	60.6	60.4	58.0	56.0	49.2	43.7	37.9
	2%	DB	48.9	47.8	53.6	63.0	72.2	73.2	81.5	81.4	77.0	65.0	54.6	47.7
		MCWB	37.1	36.3	39.8	45.3	51.8	56.9	60.4	58.0	54.3	48.0	41.9	36.2
	5%	DB	44.9	43.7	49.8	58.0	67.3	69.8	77.8	77.8	72.5	60.1	50.6	43.8
		MCWB	35.1	33.9	38.1	42.6	50.1	55.4	59.7	57.3	53.5	45.7	39.5	34.4
	10%	DB	41.9	40.3	45.9	53.2	63.0	66.5	74.2	74.0	68.0	55.8	46.8	40.7
		MCWB	33.5	31.7	36.0	40.2	47.9	53.4	58.7	56.7	51.8	43.3	37.2	32.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.2	40.0	43.9	48.8	55.6	62.9	66.7	63.4	58.9	52.0	46.0	40.1
		MCDB	51.8	50.1	56.1	66.3	71.9	74.3	77.9	76.0	74.0	65.9	56.9	50.5
	2%	WB	37.8	36.8	41.0	45.8	53.1	59.2	63.8	61.3	56.4	49.0	42.9	37.4
		MCDB	46.7	46.7	51.8	60.4	68.4	70.1	74.9	74.6	73.0	62.4	53.7	46.4
	5%	WB	35.8	34.5	38.6	43.5	51.0	56.7	62.0	59.5	54.3	46.3	40.1	34.9
		MCDB	43.9	43.2	48.8	56.4	64.9	67.4	73.2	72.9	69.9	58.8	49.4	43.2
	10%	WB	33.8	32.1	36.3	41.0	48.8	54.5	60.1	57.8	52.3	44.0	37.4	32.5
		MCDB	41.7	39.8	45.5	52.3	61.2	64.3	71.6	70.6	66.8	55.2	46.3	40.6
Mean Daily Temperature Range	5% DB	MDBR	16.4	16.3	16.3	17.1	17.7	16.8	19.3	19.7	18.3	16.6	14.9	15.1
		MCDBR	16.6	19.4	19.8	23.1	22.8	20.4	23.0	24.2	23.1	20.8	18.4	17.2
		MCWBR	10.5	12.0	11.4	12.1	11.5	11.6	11.7	10.9	10.8	10.8	10.9	11.2
	5% WB	MCDBR	15.7	19.6	19.5	22.0	20.5	18.6	19.9	20.9	21.4	19.8	18.3	17.7
		MCWBR	10.5	12.7	11.8	12.0	10.9	11.5	12.3	11.0	10.8	10.8	11.4	12.1
Clear-Sky Solar Irradiance	taub	0.241	0.253	0.272	0.319	0.325	0.323	0.324	0.338	0.301	0.269	0.247	0.225	
	taud	2.456	2.471	2.482	2.363	2.411	2.458	2.466	2.419	2.506	2.570	2.522	2.463	
	Ebn at Noon	266	290	301	294	296	296	294	284	286	280	262	258	
	Edh at Noon	18	23	28	35	35	34	33	33	27	21	17	15	
All-Sky Solar Radiation	RadAvg	375	685	1069	1497	1769	1926	2149	1738	1224	755	405	290	
	RadStd	30	56	84	103	140	140	130	116	118	80	35	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (68 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page